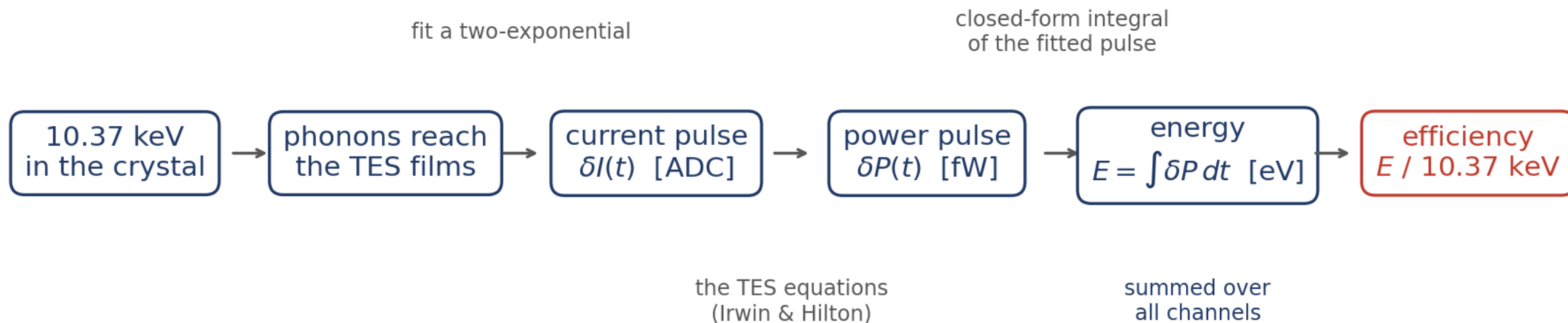


Phonon collection efficiency of Z7

Zhiheng Li · University of Minnesota · SuperCDMS SNOLAB Run 4 · September 2026
github.com/ZhihengLi0/urop → [snolab/collection_efficiency/](#)

The question



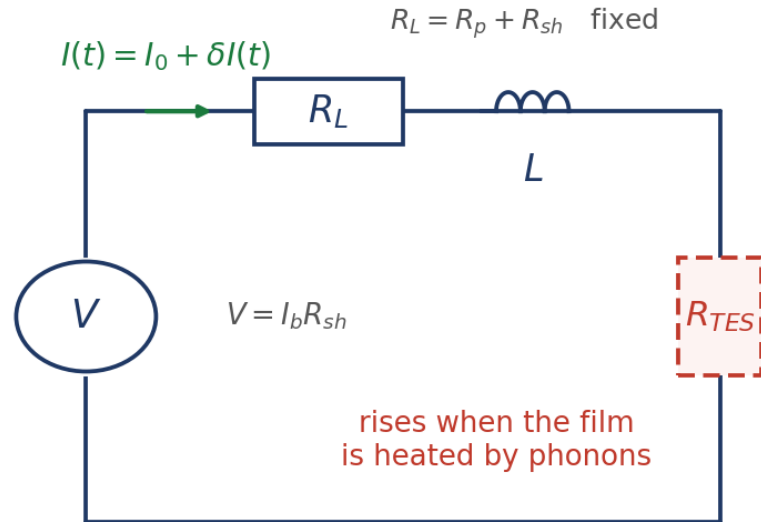
For detectors operated at 0 V (no NTL effect):

Collection efficiency = energy absorbed by the TESs, summed over all channels $\div$ the 10.37 keV of the event

Sample: Z7, the K-line events of the Ge activation run, from SNOLAB R4 (2207 events, 30 series, all 11 channels).

Where the formula comes from

the film pays for the absorbed energy out of its own Joule heating



one loop: the voltage source, the fixed load, the coil, and the TES whose resistance carries the signal

1. heat flow in the film

$$C \dot{T} = P(t) + P_J - P_{bath}$$

2. the film ends where it started, so $\int C \dot{T} dt = 0$

$$E = \int P dt = - \int \delta P_J dt$$

3. the loop gives the voltage on the TES, and its Joule power

$$V_{TES} = V - IR_L - L \dot{I}, \quad P_J = IV_{TES}$$

4. put $I = I_0 + \delta I$ and expand

$$\delta P_J = V \delta I - R_L (2I_0 \delta I + (\delta I)^2) - L I \dot{\delta I}$$

5. integrate over the pulse: the L piece cancels at the two ends, and $V = I_0(R_L + R_0)$

$$E = \int \delta P dt, \quad \delta P = I_0(R_L - R_0) \delta I + R_L (\delta I)^2$$

the note's Method 1 puts $2R_L$ in the second term, not R_L : +0.58 % in energy

An event heats the film, its resistance rises, the current falls — and the Joule heating falls by exactly what the event delivered (negative electro-thermal feedback).

So the absorbed energy can be read from the current alone: $E = -\int \delta P_J dt$, which is the formula on the next slide.

Left out: the bath leak and the inductor end terms. Both need G and L from dI/dV , which these series do not have.

The formula, and the two numbers it needs

$$\delta P(t) = I_0 (R_L - R_0) \delta I(t) + 2 R_L (\delta I(t))^2$$

linear term

quadratic term

$$E = \int \delta P(t) dt = I_0 (R_L - R_0) A (\tau_f - \tau_r) + 2 R_L A^2 \left(\frac{\tau_f + \tau_r}{2} - \frac{2 \tau_f \tau_r}{\tau_f + \tau_r} \right)$$

for the fitted pulse $\delta I(t) = A [e^{-t/\tau_f} - e^{-t/\tau_r}]$ integrated from $-\infty$ to $+\infty$: zero before it starts, decays to zero after

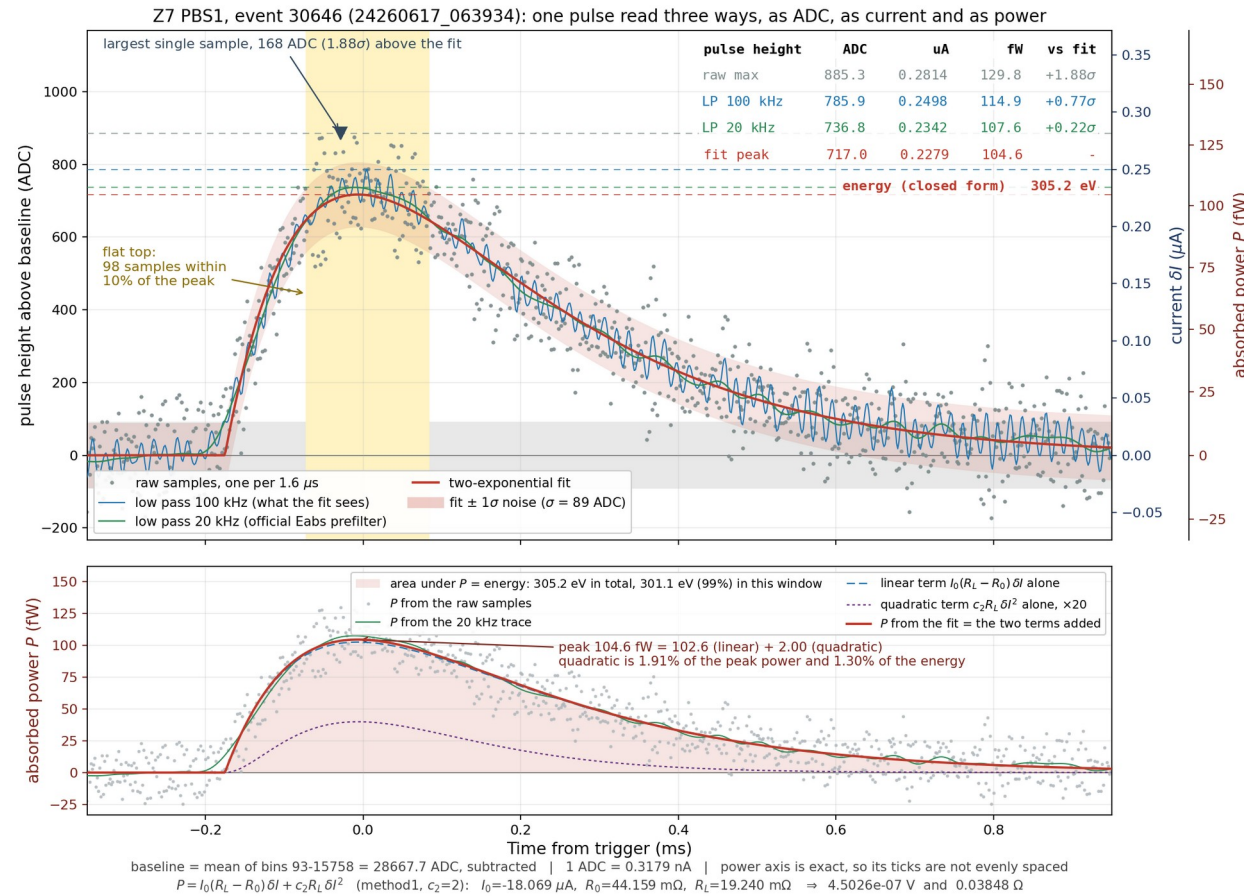
Z7 PBS1 bias point (measured, from detectorConfig): $I_0 = -18.07 \mu\text{A}$, $R_0 = 44.16 \text{ m}\Omega$, $R_L = R_p + R_{sh} = 19.24 \text{ m}\Omega$

$$\Rightarrow I_0 (R_L - R_0) = 0.45 \mu\text{V} \quad \text{and} \quad 2 R_L = 0.0385 \Omega$$

δP = the change of Joule heating when the current moves by δI : a linear term and a small quadratic term.

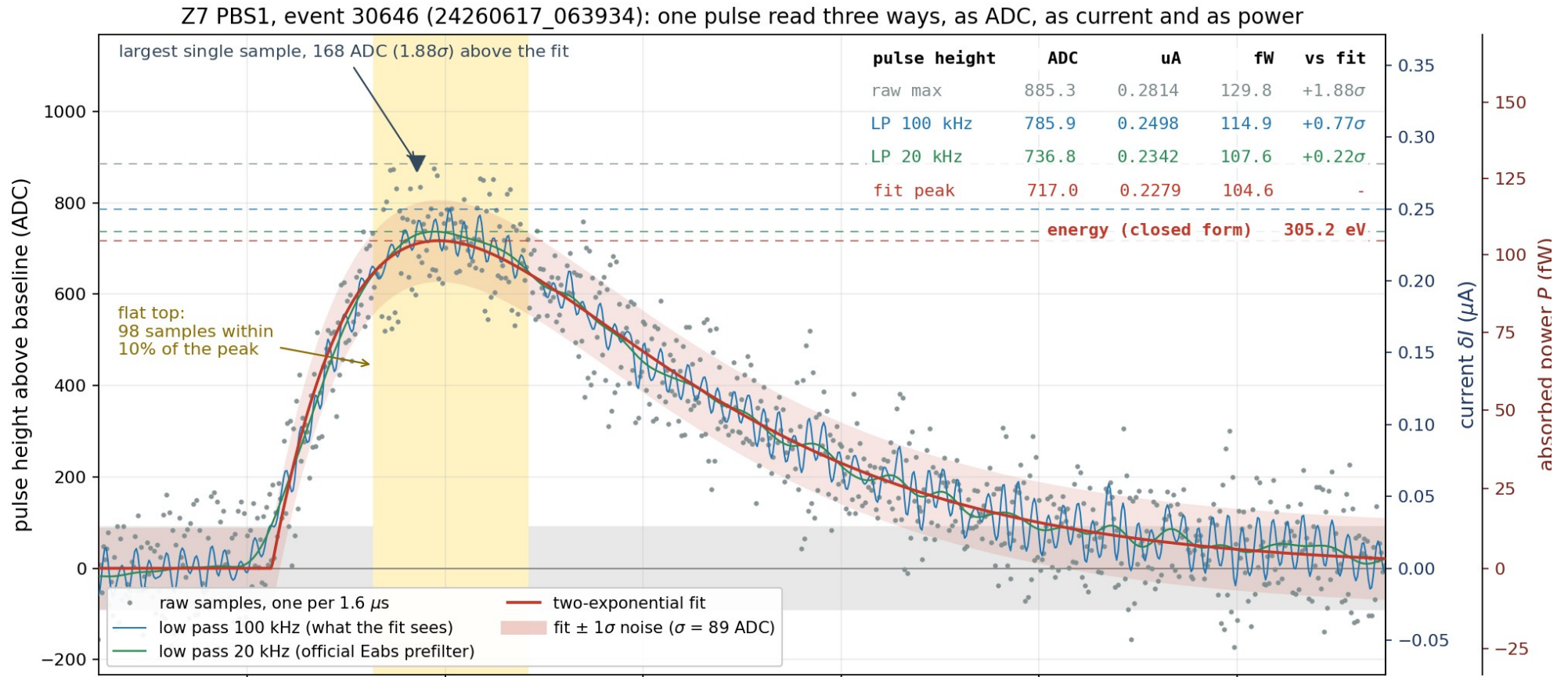
Example: one pulse

Z7 PBS1, event 30646: the same trace as ADC counts, as current, as power



The two-exponential is fitted to the 100 kHz low-passed trace (blue), not to the raw samples; the raw samples are only drawn.

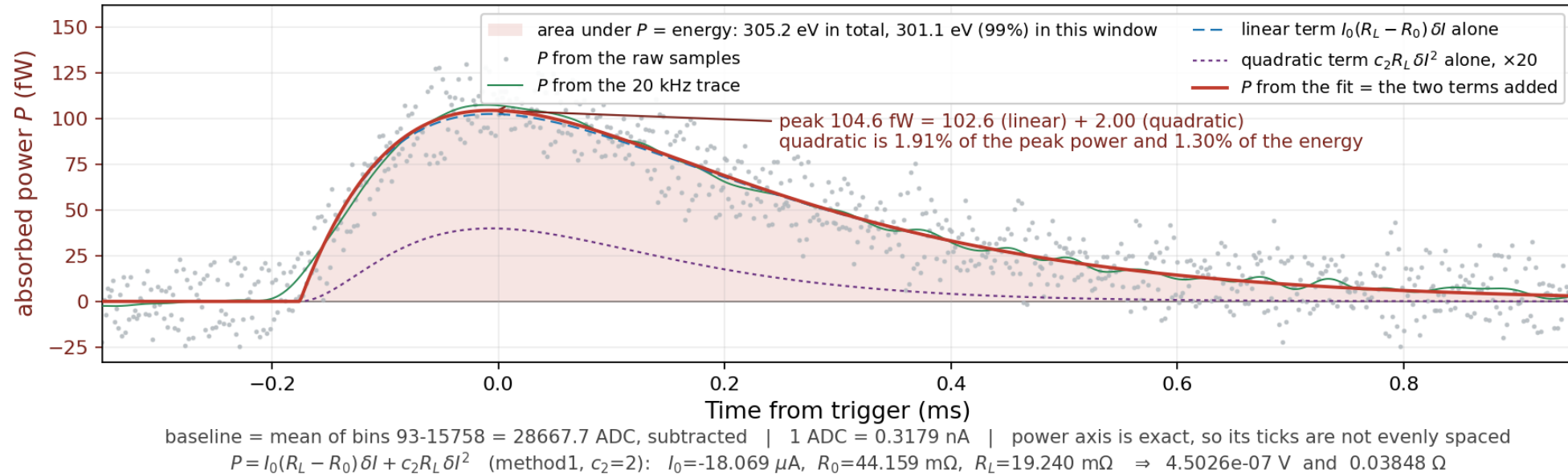
The height: take it from the fit, not from the largest sample



Yellow band: 98 samples sit within 10 % of the peak. The largest of them is the largest noise excursion.

Largest raw sample 885 ADC, fit 717 ADC: the maximum reads 23 % high. Filtering helps; the fit removes it.

The power pulse, and its area is the energy

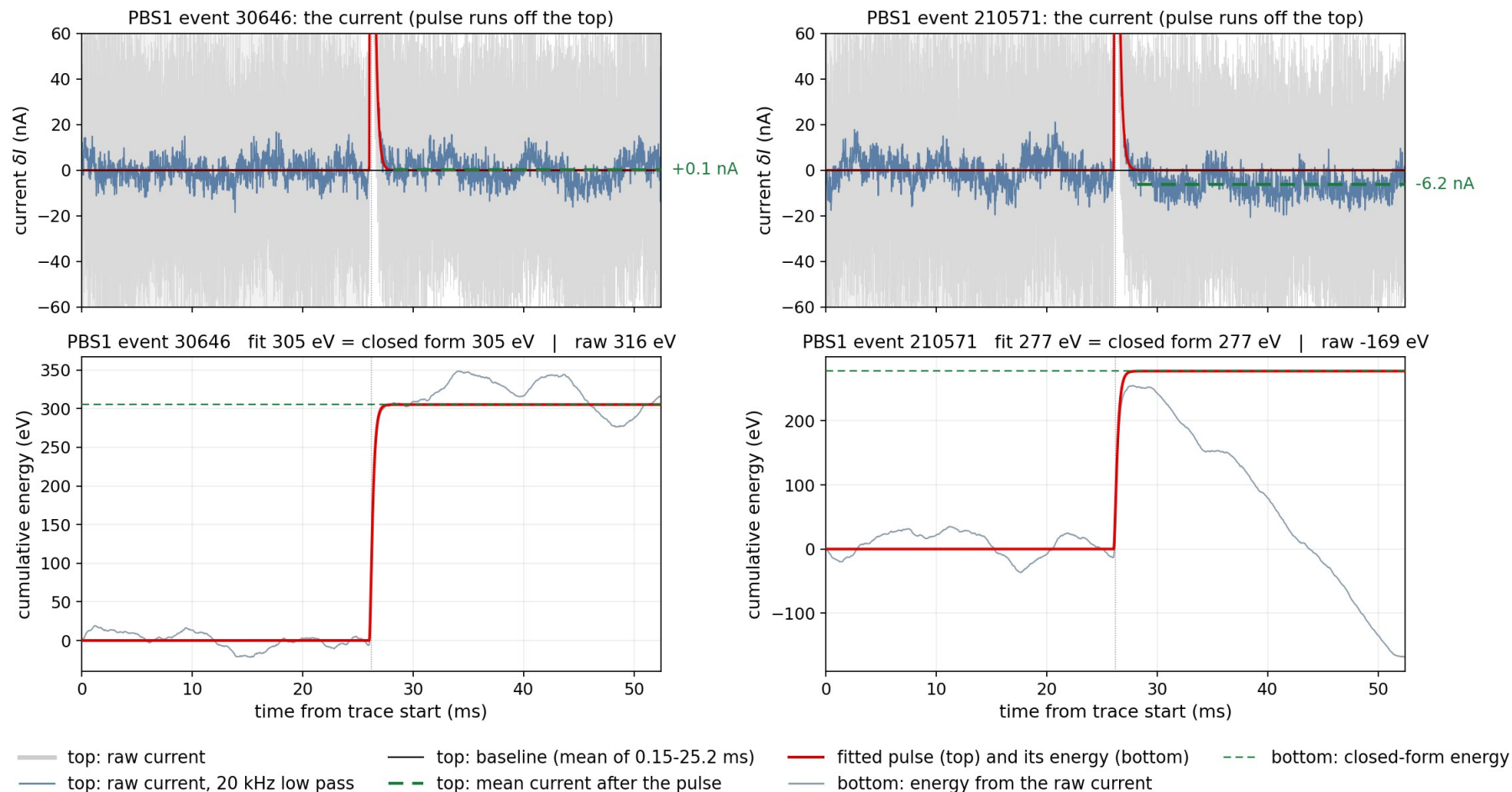


Red: the fitted current put through the formula. Blue dashed: the linear term alone. Purple: the quadratic term, $\times 20$.

The quadratic term is 1.9 % of the peak — the power pulse has the shape of the current pulse.

The shaded area is the energy this channel absorbed: 305 eV.

Why the fitted pulse is integrated, and not the data

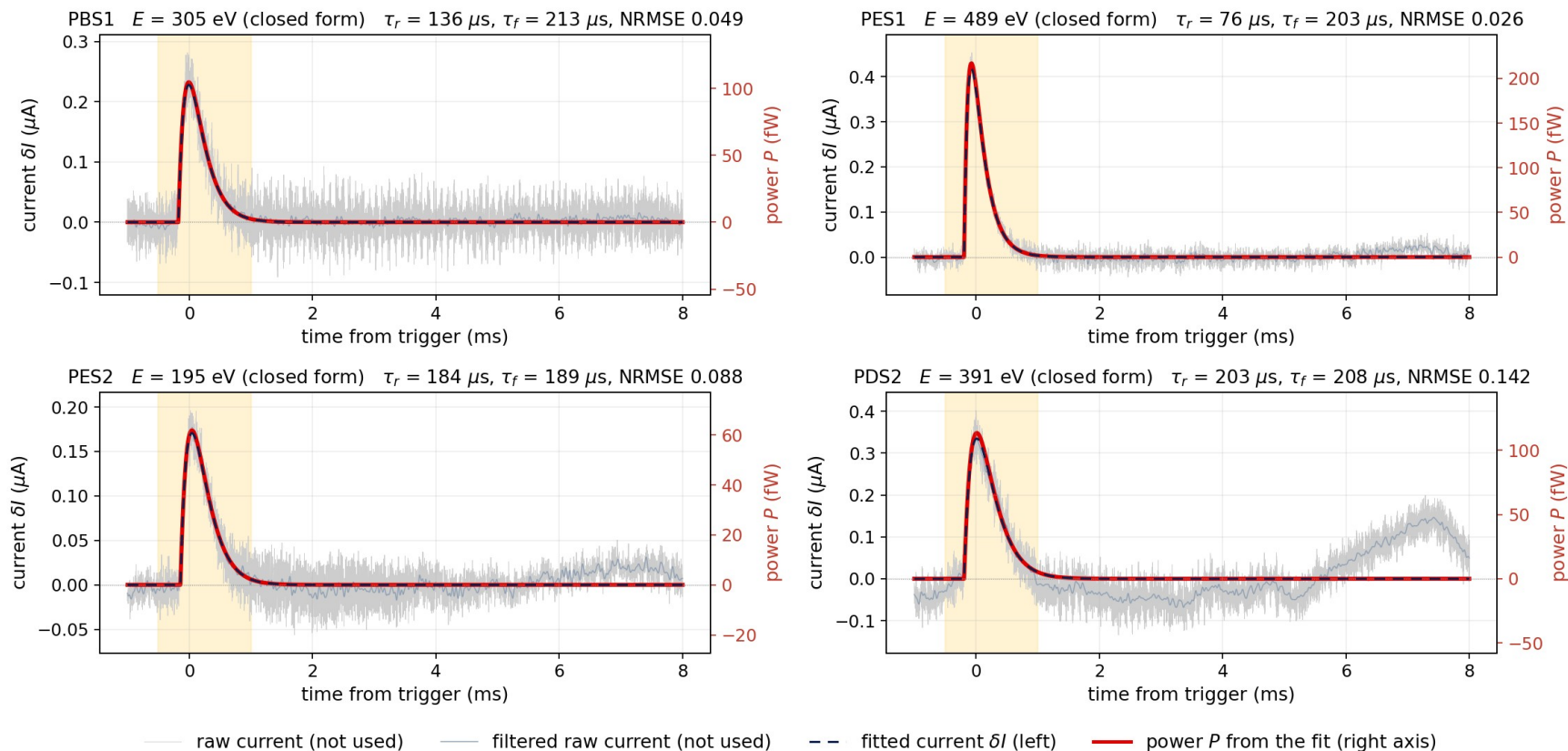


Top: the raw current of the same two events, zoomed on the baseline. Bottom: the energy accumulated from it (grey) and from the fit (red).

Right: after the pulse the current sits 6 nA below its baseline — hidden in 28 nA of noise, yet over 24 ms it is -420 eV: raw -169 eV, fit 277 eV.

The same event in its other channels: they add up to 33 %

Z7 event 30646; four of the eleven channels

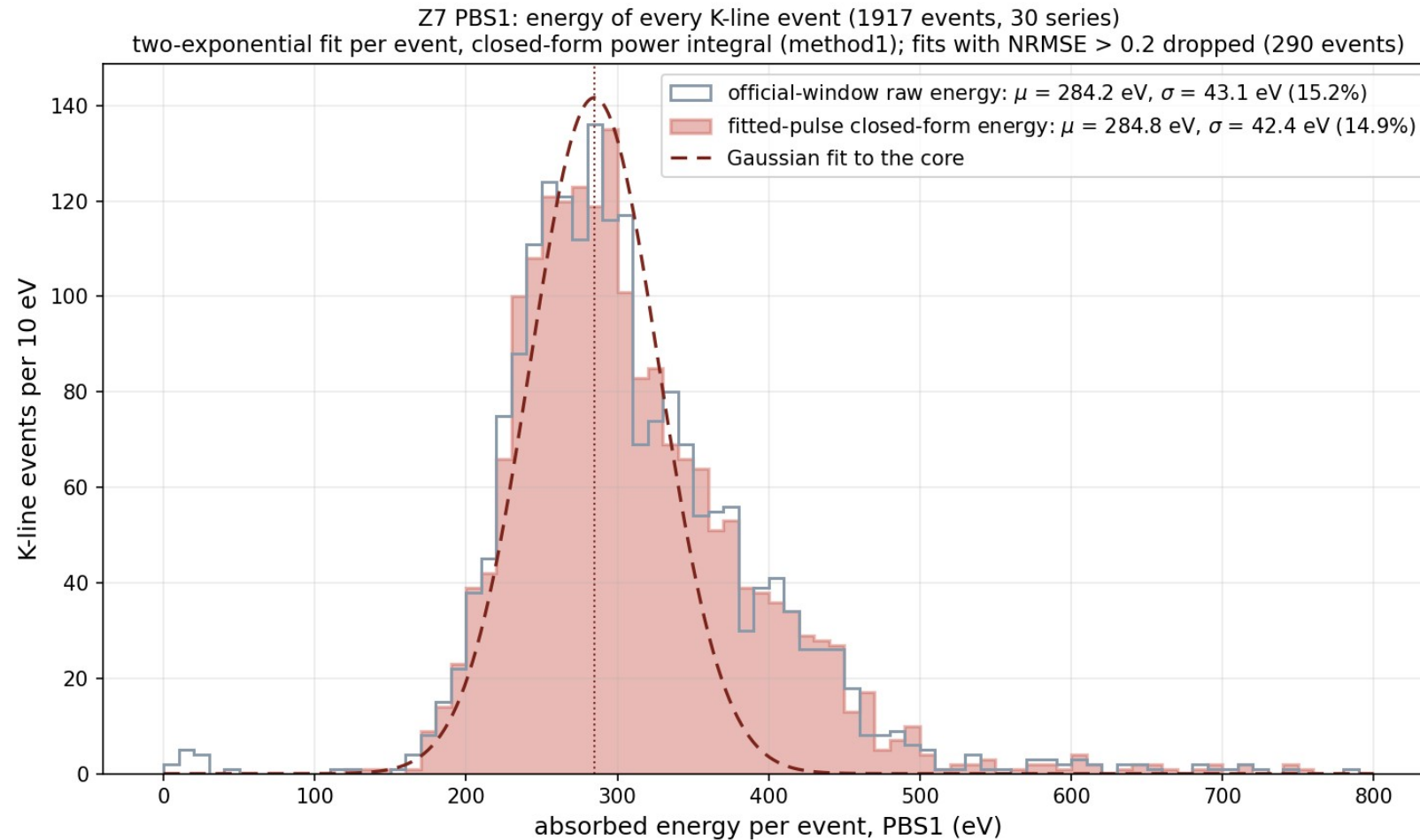


The channels share the energy unevenly (PES1 489 eV, PES2 195 eV): the event sat closer to one side.

Summed over the eleven channels: 3419 eV = 33.0 % of 10.37 keV. PDS2 (bottom right) carries a slow swing.

Now every event: one channel gives a wide, lopsided distribution

PBS1, 1917 K-line events, one fit and one integral each



Red: from the fit. Grey: the official window on the raw trace. Same peak to 0.2 %.

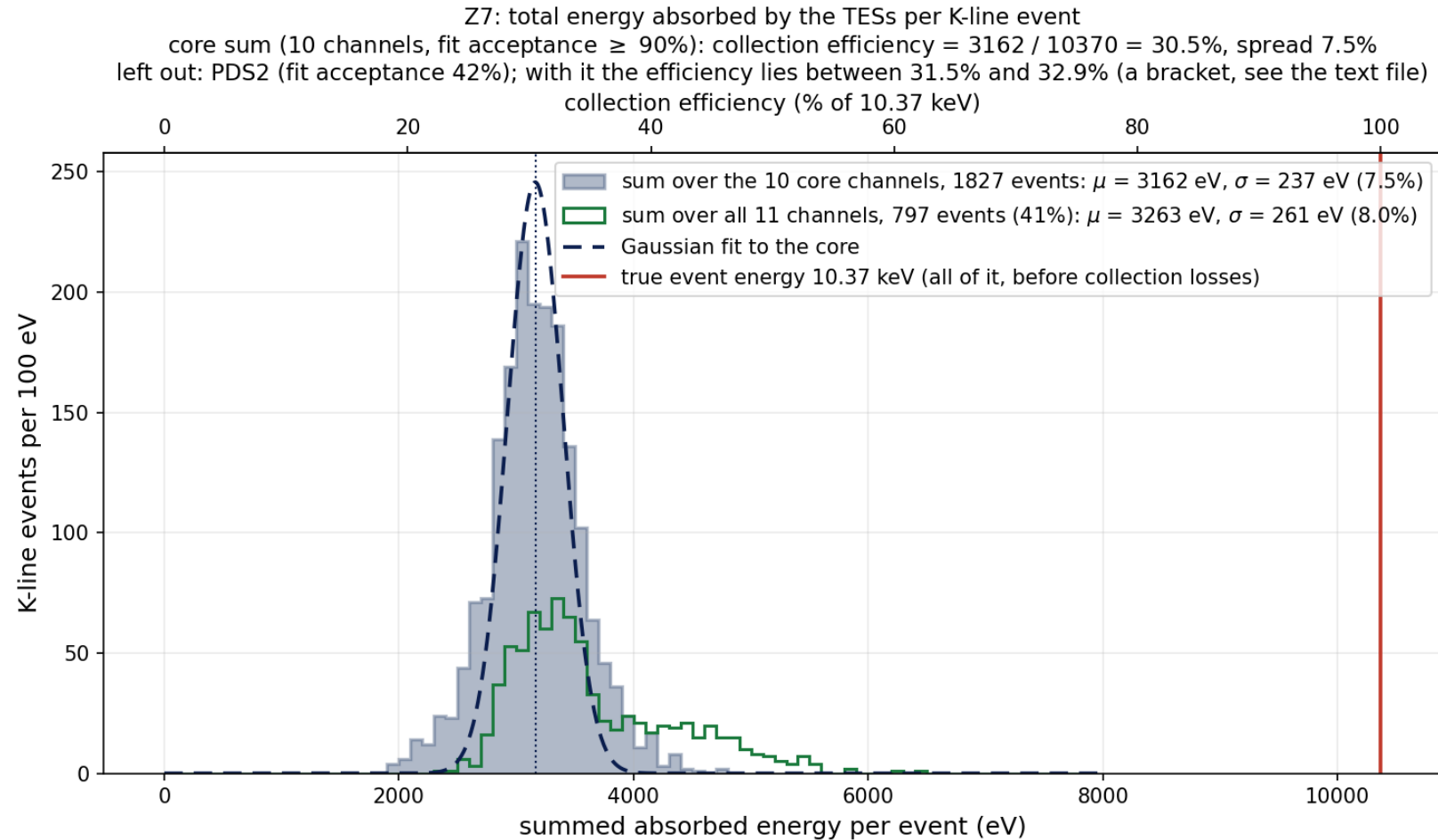
Peak 285 eV, width 15 % — for a line whose energy is fixed.

The tail to the right is the events that happened near this channel.

One channel's width is event position, not noise.

Summed over the channels: $32 \pm 1 \%$

the total energy the TESs absorbed per K-line event



Ten channels summed, 1827 events: 3162 eV = 30.5 %, and the width drops to 7.5 % — the position effect cancels.

PDS2 fits only 42 % of the events (that slow swing), and those events are not typical; its share can only be bracketed.

Full efficiency 31.5–32.9 %: $32 \pm 1 \%$.

Red line: 10.37 keV — two thirds of the energy never reaches the TESs.

Summary, and what is still open

A current pulse becomes a power pulse through the TES equations; the area under the power is the energy. The quadratic term is 2 %.

The energy comes from the fitted pulse, integrated from $-\infty$ to $+\infty$: no bias from the noisy maximum, no drift.

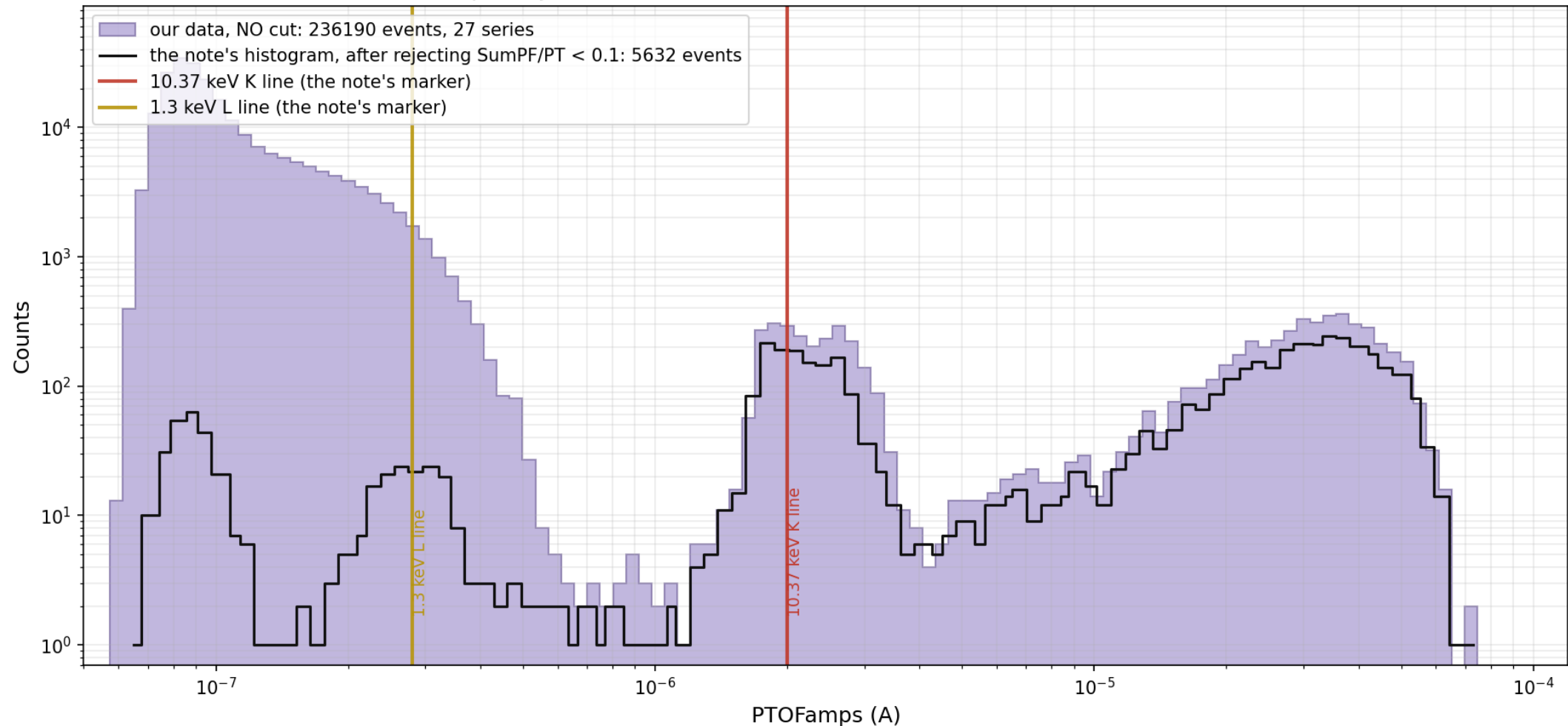
One channel: 15 % wide and lopsided (position). Ten channels summed: 7.5 % and symmetric.

Z7 absorbs 32 ± 1 % of a 10.37 keV event in its TESs. The CDMS note on the R37 CUTE tower reports 26–41 %.

Open: Z7 only · the HV bias of Z7 is not established (0 V assumed) · no dI/dV , so the inductor term is dropped · PDS2 is the ± 1 % · 12.5 % of events fit in no channel and are not in the histograms.

Backup — the events

Z7: summed PTOFamps over 27 series, rebuilt from the Prompt processing with no cut applied
the note's version of the same plot rejects SumPF/PT below 0.1; the difference between the two curves is what that cut removes



Backup — the three versions of the power formula

$\delta P = I_0(R_L - R_0) \cdot \delta I + c_2 \cdot R_L \cdot (\delta I)^2$ with $c_2 = 2$ (Method 1, used here), 1 (exact small-signal), -1 (Method 2)

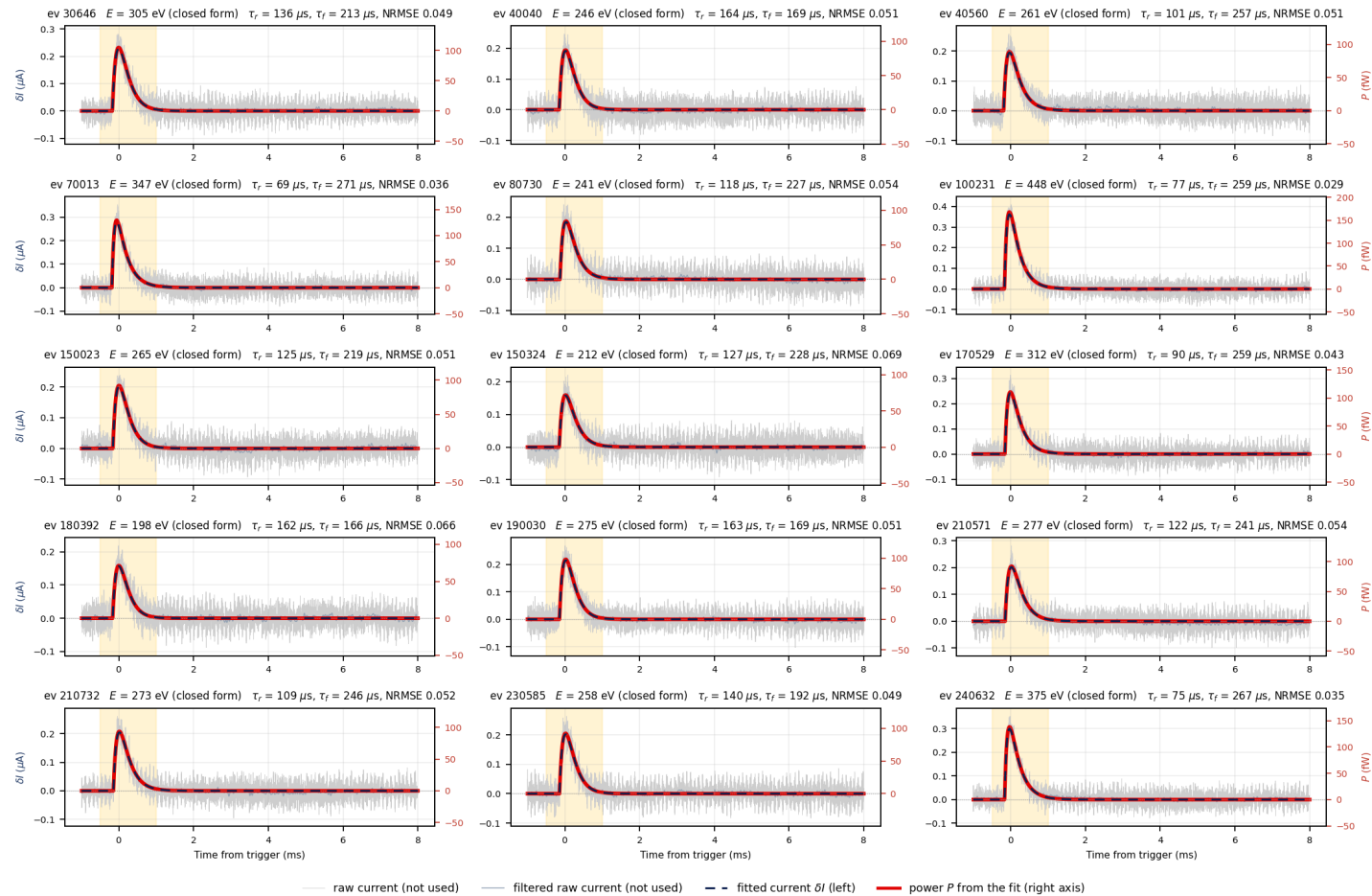
Relative to the exact form: Method 1 +0.58 %, Method 2 -1.15 % in energy — far below the 15 % single-channel width and the PDS2 bracket.

The inductor term of the full equations needs L and the loop gain from dI/dV , which do not exist for these series on MSI; it is neglected.

The official Eabs (trigger bin 16383, baseline bins 93..15758, 5-pole 20 kHz prefilter, window $-0.5/+1$ ms, same formula) is reproduced to 0.2 %; the grey outlines on the histogram slides are it.

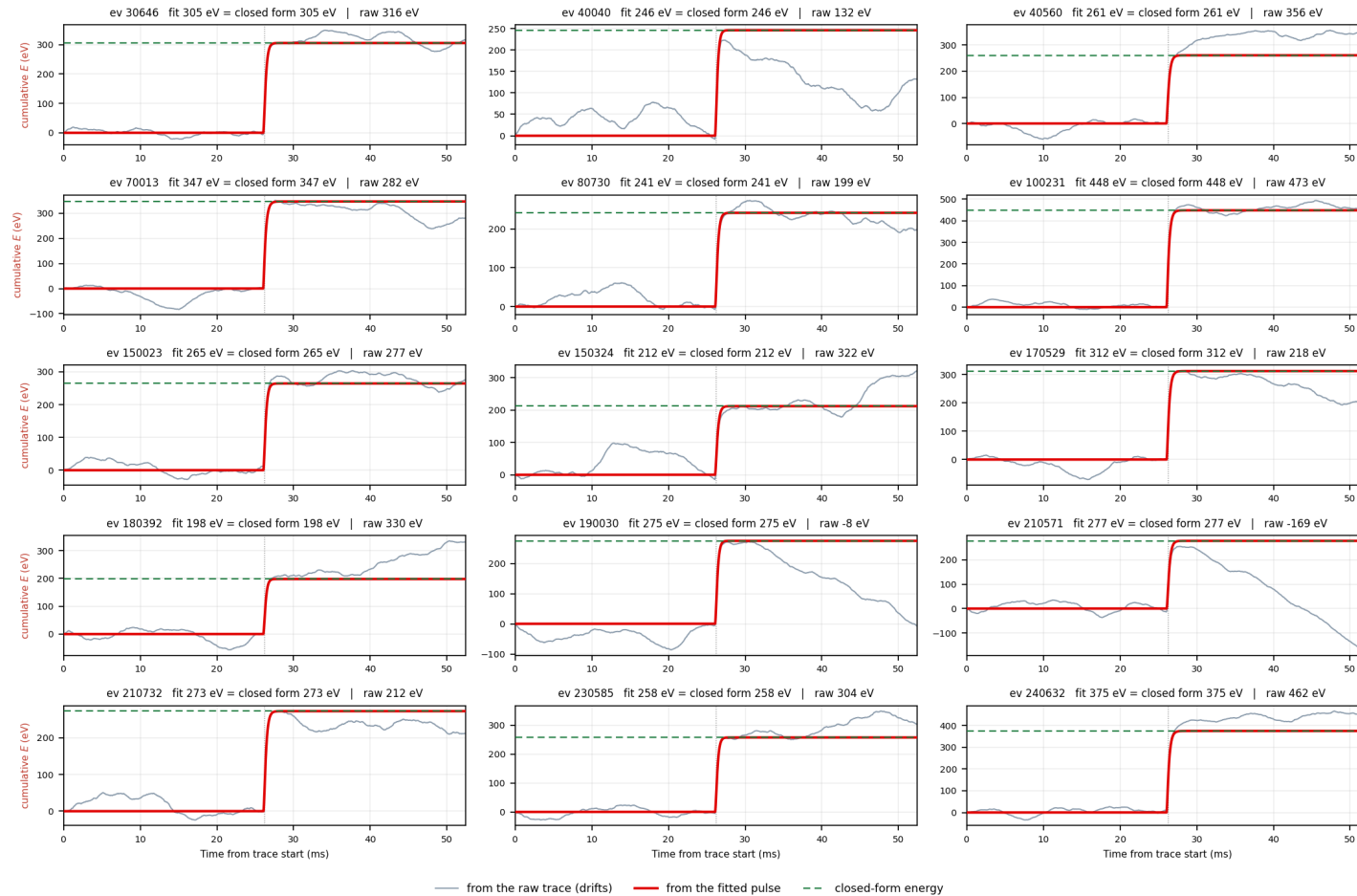
Backup — 15 events, fitted current and power (PBS1)

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)
 Z7 24260617_063934, K-line events | method1: $P(t) = 10(R/L-R_0) \cdot dI(t) + 2 \cdot R \cdot L \cdot (dI(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_r) - \exp(-t/\tau_f)]$), 5 free params, pretrigger free in 16383+-3000; the raw trace is drawn only for reference
 integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



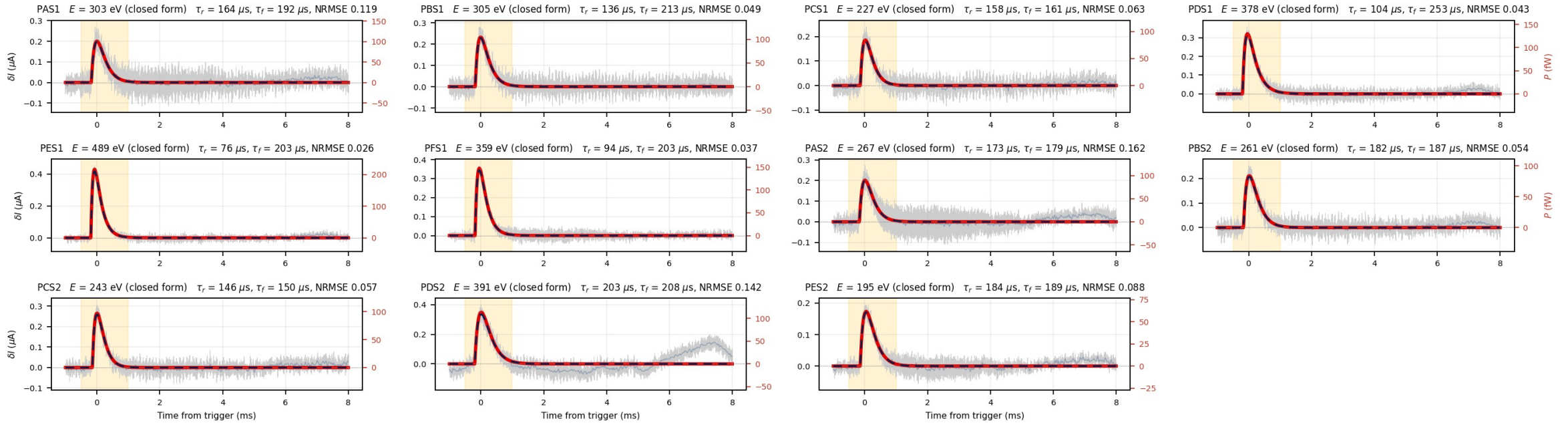
Backup — 15 events, cumulative energy (PBS1)

Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace
 Z7 24260617_063934, K-line events | method1: $P(t) = 10(R_L R_0) \cdot d(t) + 2 \cdot R_L \cdot L \cdot (d(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$), 5 free params, pretrigger free in 16383+-3000; the raw trace is drawn only for reference
 integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — event 30646, all channels

Z7 event 30646, all channels: fitted current and power
 sum over channels from the fits: 3419 eV = 33.0% of 10.37 keV (official-window raw sum 3397 eV = 32.8%)
 Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_{L-R0}) * dl(t) + 2 * R_{L-R0} * L * (dl(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$, 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
 integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — energy distribution of every channel

Z7: energy of every K-line event, one panel per channel (30 series)

each entry is one event: its trace fitted with the two-exponential and the power integrated in closed form, method1 quadratic

the dashed curve is a Gaussian fit to the core, so its centre is the PEAK; these distributions are right skewed, so peaks must not be added (the plain means do add, to 3710 eV = 35.8% of the 10.37 keV K line)

